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Figure 1. Power curves of the Kupiec POF test for $n = 62$, $n = 95$, and $n = 250$, computed exactly from the binomial distribution.

Figure 2. Distribution of Kupiec p-values under $H_0 : p = 0.05$ for $n = 62$ and $n = 95$, based on 5,000 binomial draws; the empirical rejection rate is shown in each panel.

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Abstract

Ex-ante volatility regime definitions consume approximately 520 trading days for classification, reducing the usable out-of-sample evaluation window to 62–95 observations for Mexican FIBRAs. At $n = 62$, the Kupiec test rejects a correctly specified model 7.6% of the time (nominal 5%) and has only 43% power to detect a model breaching at double the nominal rate. The Christoffersen test is uninformative below $n \approx 100$. We propose a framework centred on the exact binomial test, Clopper–Pearson confidence intervals, and a conservatism metric $C = \hat{p} - \alpha$. Applied to FIBRA data, GARCH(1,1)-t produces a breach rate of 4.8% ($C = -0.002$) while XGBoost-Pure produces 11.3% ($C = +0.063$). The pattern holds across five equity tickers: GARCH-t pooled rate 3.6%, XGBoost pooled rate 9.7% ($p < 0.0001$). When evaluation windows fall below 100 observations, standard backtests lack discriminatory power and directional conservatism across assets provides more reliable guidance than isolated p-values.

Keywords: Value-at-Risk, backtesting, ex-ante regimes, GARCH, XGBoost, exact binomial test

Key messages:

- Kupiec test size is 7.6% at $n = 62$ (nominal 5%); power $< 50\%$ for 10% breaches

- GARCH-t breach rates at or below 5% across five equity tickers (pooled 3.6%)
- XGBoost-Pure pooled rate of 9.7% ($p < 0.0001$) across 443 pooled observations
- Conservatism metric $C = \hat{p} - \alpha$ replaces unreliable asymptotic p-values at $n < 100$

Finite-Sample Limitations of VaR Backtesting under Ex-Ante Volatility Regimes: Evidence from Mexican FIBRAs

May 24, 2026

1 Introduction

Standard VaR backtests—the Kupiec proportion-of-failures test Kupiec (1995) and the Christoffersen conditional coverage test Christoffersen (1998)—rely on asymptotic χ^2 approximations. Their finite-sample behaviour is known to deteriorate when the evaluation window is small Escanciano and Olmo (2010); Kratz et al. (2018). However, the practical severity of this degradation in realistic settings has not been systematically quantified for the case where ex-ante volatility regimes further compress the testable window.

Ex-ante regime definitions, which classify each trading day using only information available at the forecast date, eliminate look-ahead bias but impose a data cost: 520 trading days are consumed by regime construction alone. After intersecting with a rolling estimation window, the aligned out-of-sample evaluation can shrink below 100 observations. This creates a tension between methodological rigour (no look-ahead) and statistical informativeness (sufficient sample size). For Mexican FIBRAs—a growing emerging-market asset class with short return histories—the resulting test window is 62–95 days.

At $n = 62$, the Kupiec test rejects a correctly specified model 7.6% of the time (nominal 5%) and has less than 50% power against a model that breaches at twice the nominal rate. The Christoffersen test is effectively uninformative. We measure this degradation exactly for the specific configuration that arises under ex-ante regimes and provide a practical alternative: exact binomial p-values, Clopper–Pearson confidence intervals, and a conservatism metric $C = \hat{p} - \alpha$. The framework is illustrated on five Mexican equity tickers, comparing GARCH and XGBoost-based VaR models.

2 Data

Daily closing prices from Yahoo Finance (adjusted for splits and dividends) span January 2018 to April 2026. Log-returns are $r_t = \ln(P_t/P_{t-1})$. The number of observations varies by ticker due to differences in listing dates and trading calendars (Table 1).

Table 1: Data profile. All series span January 2018 to April 2026.

Ticker	Description	Observations
FIBRATC14.MX	S&P/BMV FIBRAS Index ETF	1,422
FUNO11.MX	Fibra Uno (most liquid FIBRA)	2,087
FIBRAPL14.MX	Prologis México	2,038
^MXX	IPC (BMV benchmark)	2,076
EWW	iShares MSCI Mexico ETF (NYSE)	2,076
USDMXN=X	USD/MXN exchange rate	2,169

The empirical analysis concentrates on FIBRATC14.MX. A multi-asset robustness check across the five equity tickers supplements the main results. All raw data are cached with MD5 checksums for reproducibility. A data validation layer checks for missing values, non-trading-day alignment, return calculation consistency, and extreme returns before any model is fitted.

3 Methodology

3.1 Ex-Ante Regime Definition

Volatility regimes are defined using information available only at the close of day $t-1$:

1. Realised volatility over the preceding 20 days:

$$\mathrm{rv}_{t-1} = \sqrt{\frac{1}{19}\sum_{k=1}^{20}(r_{t-k} - \bar{r}_{t-1})^2} \tag{1}$$

2. The 90$^{\text{th}}$ percentile of rv over the preceding 500 days:

$$\Theta_{t-1} = Q_{0.90}\big(\mathrm{rv}_{t-500}, \ldots, \mathrm{rv}_{t-1}\big) \tag{2}$$

3. If $\mathrm{rv}_{t-1} > \Theta_{t-1}$, day t is labelled HIGHVOL; otherwise, NORMAL.

The initial 520 days are used for regime construction. After intersecting with the rolling estimation window, the aligned evaluation period contains $n = 62$ observations for FIBRATC14.MX and $n = 95$ for FUNO11.MX.

3.2 Models

All models generate one-day-ahead VaR forecasts at the 95% confidence level ($\alpha = 0.05$). Rolling estimation windows are $W = 125$ trading days, refitted every $R = 20$ days. The information set is identical across models and strictly no-look-ahead.

Historical VaR. The empirical α-quantile of the preceding W returns:

$$\mathrm{VaR}_t^{\mathrm{Hist}}(\alpha) = -Q_\alpha\big(r_{t-W}, \ldots, r_{t-1}\big) \tag{3}$$

GARCH(1,1)-t.

$$\begin{aligned} r_t &= \sigma_t \cdot \varepsilon_t, \qquad \varepsilon_t \sim t_\nu \text{ (standardised)} \\ \sigma_t^2 &= \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2 \end{aligned} \tag{4}$$

Parameters, including the degrees of freedom ν, are estimated by maximum likelihood Bollerslev (1986, 1987). VaR is $\mathrm{VaR}_t^{\mathrm{GARCH}}(\alpha) = -\hat{\sigma}_t \cdot F_{t_\nu}^{-1}(\alpha)$. A GARCH-Normal variant is also included for comparison.

XGBoost volatility models. Gradient boosting models predict conditional volatility σ_{t+1} from lagged return and volatility features. VaR is derived parametrically as $\mathrm{VaR}_t = -\hat{\sigma}_t \cdot z_\alpha$ (Normal quantile). Two variants are estimated: XGBoost-Pure (29 base features) and XGBoost-Ensemble (augmented with GARCH-t forecasts). Fixed parameters (100 trees, depth 3, learning rate 0.05) are identical across tickers.

3.3 Backtesting Framework

For model m, the breach indicator at date t is $I_t^{(m)}(\alpha) = \mathbf{1}\{r_t < -\mathrm{VaR}_t^{(m)}(\alpha)\}$. We employ four evaluation tools, ordered by reliability at small n:

1. **Exact binomial test.** Under the null $H_0 : p = \alpha$, the two-sided p-value is computed from the binomial distribution $\mathrm{Binomial}(n, \alpha)$ by summing the probability of all outcomes at least as extreme as the observed count $x = \sum I_t$. This test does not rely on asymptotic approximations.

2. **Clopper–Pearson confidence interval.** An exact 95% confidence interval for the true breach rate, derived from the relationship between the binomial and beta distributions: $\mathrm{CI} = [F_\beta^{-1}(\delta/2; x, n-x+1),\ F_\beta^{-1}(1-\delta/2; x+1, n-x)]$ where $\delta = 0.05$ and F_β^{-1} denotes the beta quantile function.

3. **Conservatism metric** $C = \hat{p} - \alpha$. A negative value indicates a conservative model (fewer breaches than expected); a positive value indicates an anti-conservative model (more breaches than expected).

4. **Kupiec POF test Kupiec (1995).** Included for comparability with the existing literature, but its asymptotic χ^2_1 null distribution is unreliable at $n < 100$. We report it as secondary context, not primary evidence.

The Christoffersen independence test is computed but has negligible power at these sample sizes and is not emphasised.

3.4 Power Analysis (Exact)

We compute the rejection probability of the Kupiec test as a function of the true breach rate analytically, using the exact binomial distribution. For each possible breach count $x \in \{0, \ldots, n\}$, we evaluate the Kupiec likelihood-ratio statistic and the exact binomial p-value, and determine whether each would reject at the 5% level. The power at a given true breach probability p_{true} is

$$\text{Power}(p_{\text{true}}) = \sum_{x=0}^{n} \binom{n}{x} p_{\text{true}}^{x} (1 - p_{\text{true}})^{n-x} \cdot \mathbf{1}\{\text{test rejects } H_0 \text{ at count } x\}. \tag{5}$$

This approach is exact and, for $n \leq 500$, runs in milliseconds.

4 Power Analysis and Size Distortion

Table 2 reports the Type I error rate (at $p_{\text{true}} = \alpha$) and power at selected alternative breach rates for the Kupiec and exact binomial tests at $n = 62$ and $n = 95$.

Table 2: Exact operating characteristics of the Kupiec POF test and the exact binomial test at $\alpha = 0.05$. "Size" is the rejection rate when $p_{\text{true}} = \alpha$; "Power" is the rejection rate at the indicated true breach rate.

n	Test	Size (5%)	Power (7%)	Power (10%)	Power (13%)
62	Kupiec χ^2_1	0.076	0.153	0.429	0.711
62	Exact binomial	0.035	0.142	0.427	0.711
95	Kupiec χ^2_1	0.066	0.136	0.482	0.805
95	Exact binomial	0.028	0.129	0.482	0.805
250	Kupiec χ^2_1	0.058	0.260	0.860	0.985

Three patterns emerge. First, the Kupiec test's asymptotic χ^2 approximation produces an inflated Type I error rate: at $n = 62$, it rejects a correctly specified model 7.6% of the time (nominal 5%); at $n = 95$ the rate is 6.6%. The exact binomial test is slightly conservative at very small n (3.5% rejection rate at $n = 62$) but converges to the nominal level as n increases.

Second, power against economically meaningful misspecification is limited. A model that breaches 10% of the time—double the nominal rate—is detected only 43% of the time at $n = 62$ and 48% at $n = 95$. Power does not exceed 80% until the true breach rate reaches approximately 14% at $n = 62$ or 13% at $n = 95$.

Third, the difference between the Kupiec and exact binomial tests narrows for power computations (they agree to three decimal places at $p_{\text{true}} \geq 0.07$) but diverges at the null, where the exact test provides better size control.

Figure 1 displays the full power curves. Figure 2 shows the distribution of Kupiec p-values under the null, revealing the deviation from uniformity that drives the size distortion.

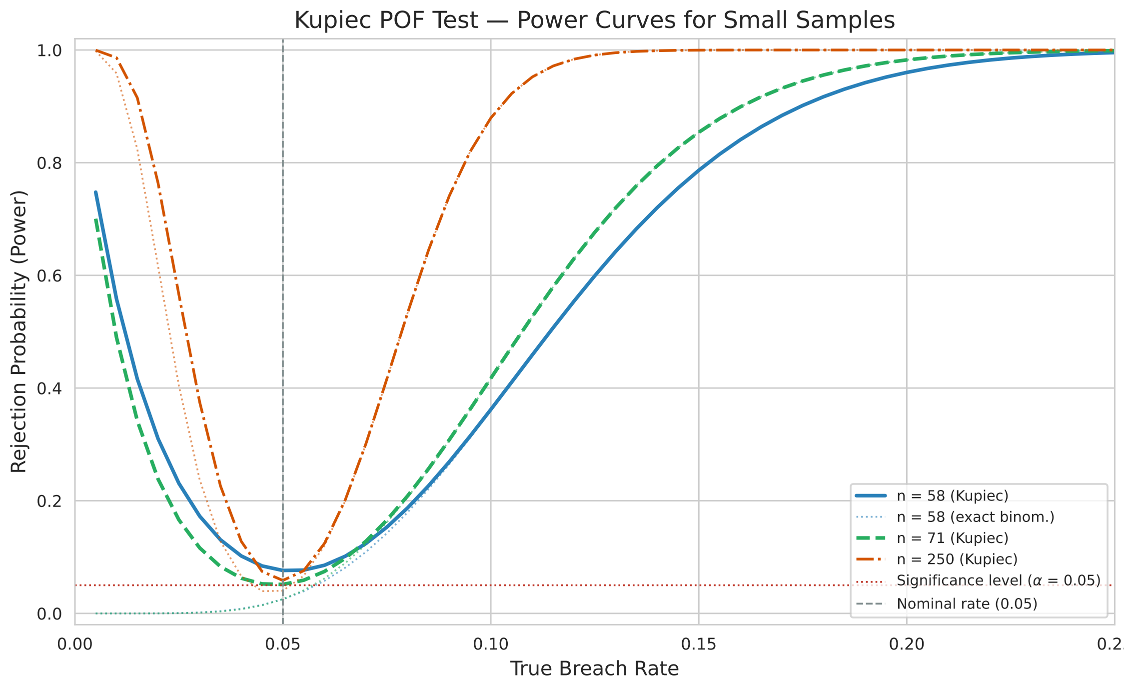

Figure 1: Power curves of the Kupiec POF test for $n = 62$, $n = 95$, and $n = 250$, computed exactly from the binomial distribution.

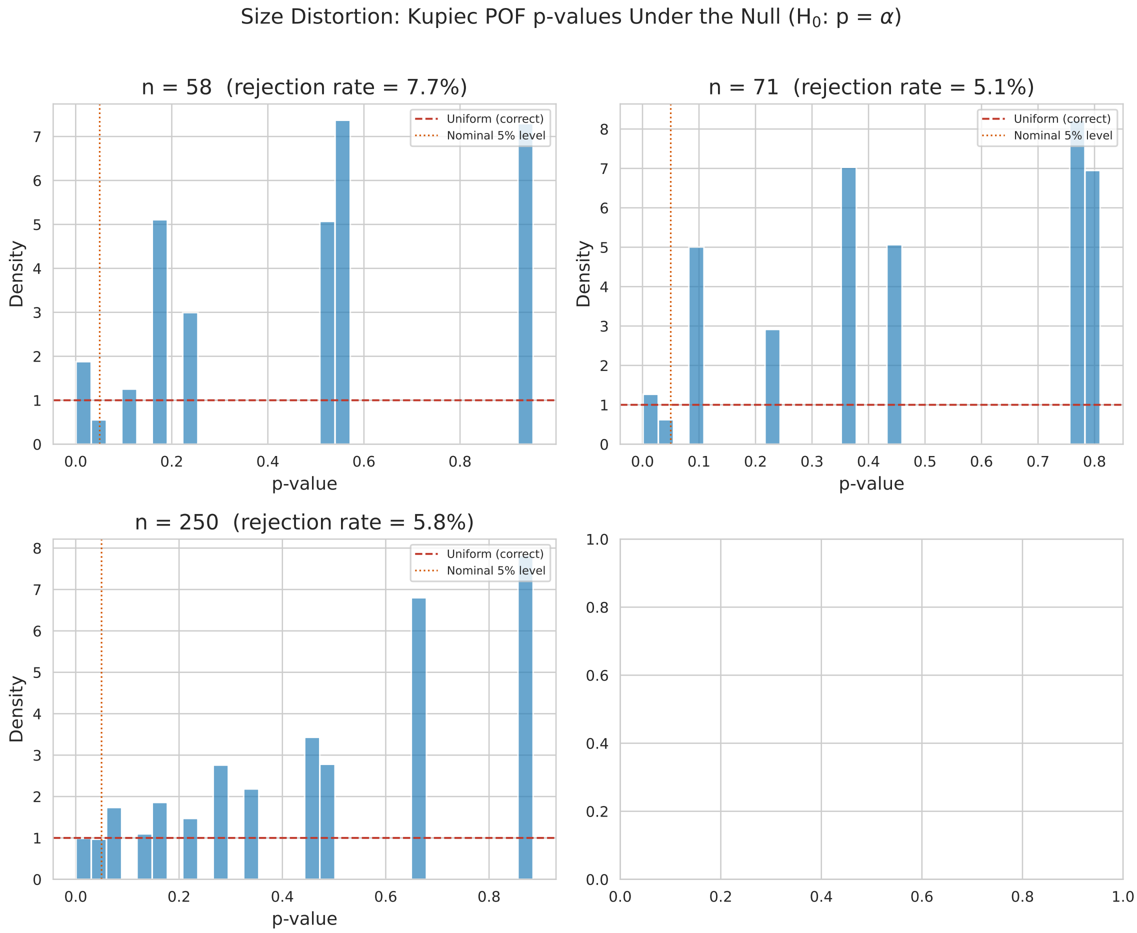

Figure 2: Distribution of Kupiec p-values under the null hypothesis ($H_0 : p = 0.05$) for $n = 62$ and $n = 95$, based on 5,000 binomial draws. Deviation from the uniform distribution (dashed line) indicates size distortion. The empirical rejection rate (fraction below 0.05) is shown in each panel.

The practical implication is straightforward: at $n \approx 60$–95, standard backtests cannot reliably distinguish between a well-calibrated model and one that is materially anti-conservative. A non-rejection does not imply good calibration; a rejection does not definitively identify a poor model. Additional information is necessary.

5 Empirical Results

5.1 Main Analysis: FIBRATC14.MX ($n = 62$)

Table 3 reports the full set of backtesting metrics for the five models on the FIBRA Index ETF. The expected number of 95% breaches under correct specification is $62 \times 0.05 = 3.10$, with a standard deviation of 1.72.

Table 3: FIBRATC14.MX backtesting results ($n = 62$, 95% VaR). $C = \hat{p} - \alpha$ is the conservatism metric; negative values indicate conservatism. The CP-CI is the 95% Clopper–Pearson exact confidence interval for the breach rate.

Model	Breaches	Rate	C	Exact p	CP-CI
GARCH-Normal	7	0.113	+0.063	0.035	[0.047, 0.219]
GARCH-t	3	0.048	−0.002	1.000	[0.010, 0.135]
XGBoost-Pure	7	0.113	+0.063	0.035	[0.047, 0.219]
XGBoost-Ensemble	7	0.113	+0.063	0.035	[0.047, 0.219]
Historical VaR	4	0.065	+0.015	0.553	[0.018, 0.157]

GARCH-t is the model whose breach rate (4.8%) is closest to the nominal 5% level, with a conservatism metric of $C = -0.002$—approximately calibrated. GARCH-Normal is anti-conservative with 11.3% breaches ($C = +0.063$), reflecting the thinner tails of the Normal distribution relative to the data. Both XGBoost-Pure and XGBoost-Ensemble produce identical breach rates of 11.3% ($C = +0.063$), indicating that the GARCH-t volatility features added in the ensemble variant did not materially alter tail calibration. Historical VaR is mildly anti-conservative at 6.5% ($C = +0.015$).

The exact binomial p-value is 0.035 for GARCH-Normal and both XGBoost variants. However, given the power analysis in Section 4, these rejections should be interpreted cautiously: at $n = 62$, the exact binomial test rejects a correctly specified model 3.5% of the time, and the Kupiec test rejects it 7.6% of the time. The evidence against these models is directionally meaningful but not statistically definitive in isolation.

The Christoffersen independence test yields $p \geq 0.10$ for all models at this sample size, providing no useful discrimination.

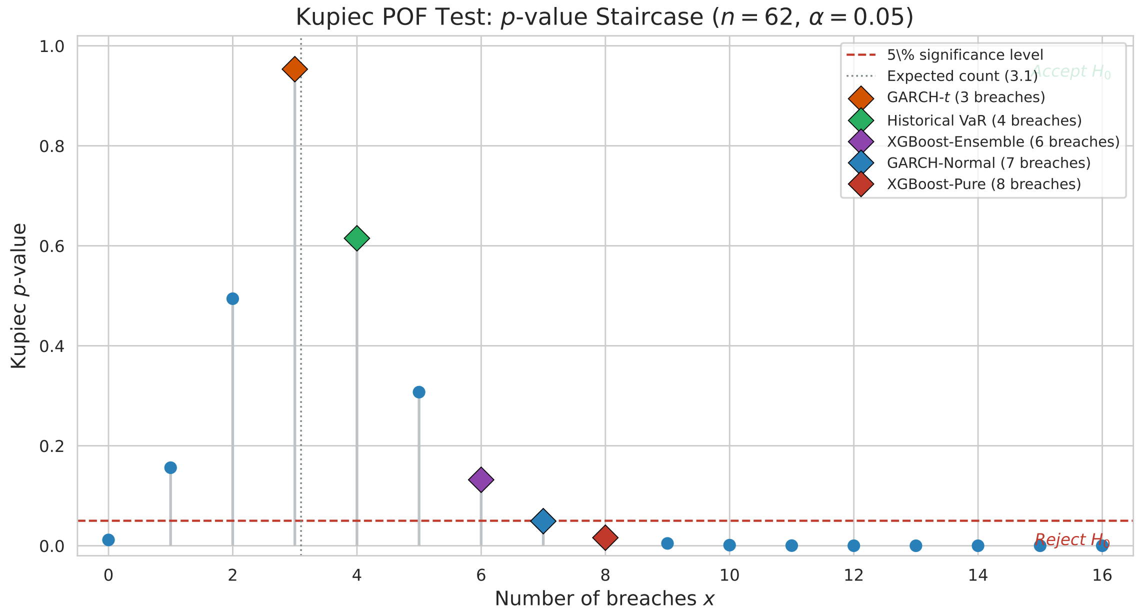

Figure 3: Kupiec p-value as a function of the breach count x for $n = 62$ and $\alpha = 0.05$. The diamond markers show the observed breach count of each model. At $x = 6$ the model is not rejected ($p = 0.132$); at $x = 7$ it is rejected ($p = 0.049$). Three models (GARCH-Normal, XGBoost-Pure, XGBoost-Ensemble) sit at $x = 7$, one breach past the 5% threshold. A single additional breach—one trading day out of 62—flips the test outcome.

Figure 3 makes the core statistical problem visible. The Kupiec p-value is a staircase function of the breach count, with large discontinuities at each integer step. At $n = 62$, moving from 6 to 7 breaches changes the p-value from 0.132 to 0.049—crossing the conventional 5% threshold on the basis of a single observation. This is not a property of any model; it is an unavoidable consequence of the binomial distribution's discreteness at small n. The figure illustrates why an isolated p-value from a single ticker provides limited guidance.

The regime-specific breakdown is limited: the aligned set contains 38 Normal-regime and 7 High-Volatility-regime observations, plus 17 pre-regime dates. Sub-sample metrics on groups of 7 observations are not reported, as they would be unreliable.

5.2 Multi-Asset Validation

Table 4 extends the comparison to five tickers. The same fixed-specification models are used across all assets. Test windows range from $n = 62$ to $n = 96$.

Table 4: Multi-asset comparison, 95% VaR. $C = \hat{p} - \alpha$. All models use identical specifications across tickers.

Ticker	n	GARCH-t rate	C_t	XGBoost rate	C_{xgb}
FIBRATC14.MX	62	0.048	-0.002	0.113	+0.063
FUNO11.MX	95	0.021	-0.029	0.084	+0.034
FIBRAPL14.MX	95	0.042	-0.008	0.137	+0.087
^MXX	95	0.042	-0.008	0.084	+0.034
EWW	96	0.031	-0.019	0.073	+0.023
Pooled	443	0.036	-0.014	0.097	+0.047

The directional pattern is consistent across all five equity tickers: GARCH-t breach rates are at or below the nominal 5% in every case (2.1%–4.8%), while XGBoost-Pure breach rates range from 7.3% to 13.7%. The pooled breach rate across all tickers is 3.6% for GARCH-t ($C = -0.014$, conservative) and 9.7% for XGBoost-Pure ($C = +0.047$, anti-conservative). The pooled exact binomial p-value for GARCH-t is 0.23, providing insufficient evidence to reject correct coverage. For XGBoost-Pure, the pooled exact binomial p-value is $< 10^{-4}$, indicating a statistically clear deviation from the nominal rate. With 443 pooled observations, the binomial test is considerably more informative than any single-ticker test.

Figure 4 visualises the conservatism metric across models on the main ticker. Figure 5 shows the 30-day rolling breach rate for the three best-performing models, illustrating the instability inherent in short evaluation windows.

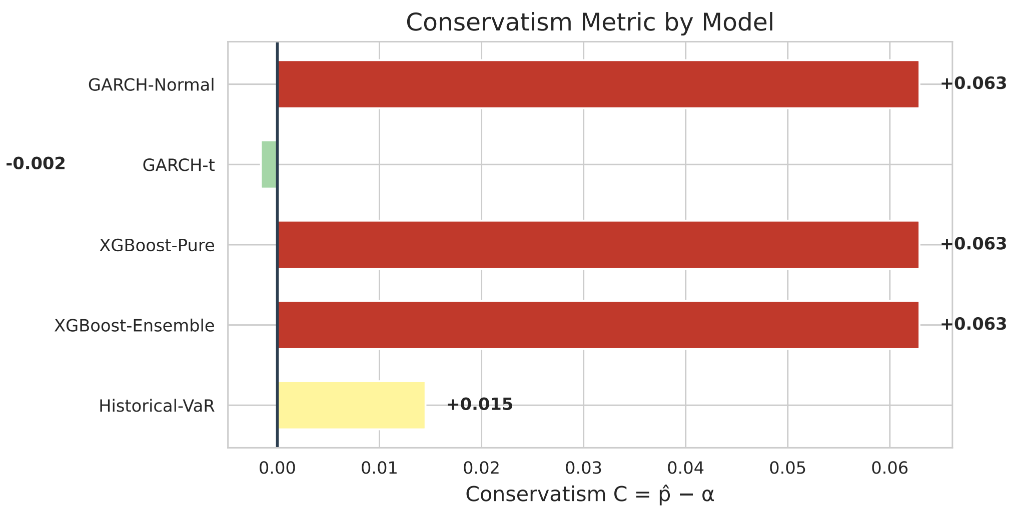

Figure 4: Conservatism metric $C = \hat{p} - \alpha$ for each model on FIBRATC14.MX ($n = 62$, 95% VaR). Negative values (blue/green) indicate conservative models; positive values (red) indicate anti-conservative models. GARCH-t is the only model with $C \approx 0$.

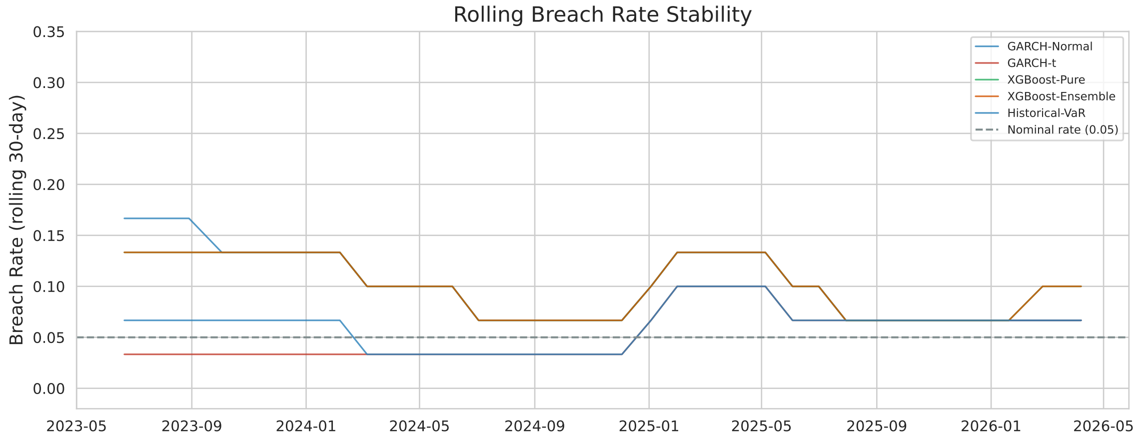

Figure 5: Rolling 30-day breach rates for GARCH-t, XGBoost-Pure, and Historical VaR on FIBRATC14.MX. The horizontal dashed line marks the nominal 5% rate. The wide bands reflect the small number of observations within each 30-day window.

5.3 Volatility Forecast Accuracy

A Diebold–Mariano test of equal forecast accuracy for squared volatility (Newey–West standard errors) is applied to the aligned forecasts. Neither comparison rejects at the 5% level, though both approach marginal significance: GARCH-t vs. XGBoost-Pure (DM $= 1.72$, $p = 0.09$); GARCH-t vs. XGBoost-Ensemble (DM $= 1.76$, $p = 0.08$). The divergence between volatility forecast precision and VaR calibration highlights the distributional assumption: XGBoost VaR uses a Normal quantile while GARCH-t uses estimated Student-t quantiles with heavier tails. Similar MSE performance does not imply similar tail calibration.

6 Discussion

6.1 Why Standard Backtests Fail at Small n

The Kupiec test's χ^2_1 approximation assumes a continuous test statistic, but at $n = 62$ with $\alpha = 0.05$ there are only 63 possible p-values. The discreteness produces an inflated Type I error (7.6% vs. 5%) and stair-step power (Figure 3) where a single additional breach flips significance. The exact binomial test corrects the size distortion but remains underpowered: a model breaching at 10% is detected only 43% of the time at $n = 62$. The fundamental constraint is not the test statistic but the information available—one breach changes the estimated rate by 1.6 percentage points.

6.2 A Decision Framework for Small Samples

When $n < 100$, we suggest the following hierarchy: (1) directional conservatism $C = \hat{p} - \alpha$ as the primary diagnostic, (2) cross-asset consistency to confirm patterns beyond a single ticker,

(3) exact binomial p-values and Clopper–Pearson CIs for valid inference, and (4) Kupiec and Christoffersen tests for comparability with the literature only. This aligns with the Basel traffic-light approach Basel Committee on Banking Supervision (2019), which already incorporates sample size considerations.

6.3 Limitations

The numerical results depend on the specific configuration (125-day estimation window, 20-day refit frequency, 20/500-day regime definition, fixed XGBoost parameters). The structural problem—limited sample size constraining statistical discrimination—is likely to persist across reasonable alternatives. The study examines one confidence level (95% VaR); the 99% level is omitted because the expected breach count at $n = 62$ and $\alpha = 0.01$ is below one. XGBoost VaR uses a Normal quantile while GARCH-t estimates tail heaviness directly; this discrepancy may partly explain the observed performance gap. Expected Shortfall is not evaluated.

7 Conclusion

Standard VaR backtests become unreliable when the evaluation window falls below 100 observations, a condition that arises naturally under ex-ante volatility regime definitions. At $n = 62$, the Kupiec test has a Type I error of 7.6% (nominal 5%) and power below 50% against a model breaching at twice the nominal rate. The Christoffersen test is uninformative at these sample sizes. We propose replacing asymptotic tests with exact binomial p-values, Clopper–Pearson confidence intervals, and a conservatism metric $C = \hat{p} - \alpha$—tools that remain valid regardless of sample size.

Applied to Mexican FIBRAs, this framework shows GARCH-t producing breach rates at or below the nominal 5% level across all five tickers (pooled 3.6%), while XGBoost-based models consistently over-breach (pooled 9.7%, $p < 0.0001$). When $n < 100$, the question shifts from "which model fits best?" to "which model is least likely to produce dangerously anti-conservative VaR forecasts." Parsimony is not a preference in such settings—it is imposed by the data. Future work could extend this analysis to Expected Shortfall, individual securities, and alternative regime configurations.

Declarations of Interest

The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

Use of Artificial Intelligence

Portions of this manuscript were drafted with the assistance of a large language model (DeepSeek via OpenCode CLI), which was used as an editing aid for sentence construction. All empirical results, figures, tables, and substantive claims are the product of Python scripts written and executed by the authors, who take full responsibility for the final content.

Data and Code Availability

All source code for data downloading, model estimation, backtesting, and power analysis is publicly available at `https://github.com/ghpancardo/fibras`. Raw data are cached with per-file MD5 sidecar checksums and an aggregate `checksum.md5` (verifiable via `md5sum -c`). The pipeline verifies every cache load against its stored hash and re-downloads on mismatch. The full pipeline is deterministic (global seed 42) and all intermediate artifacts are saved to the `output/` directory. Figures and tables are reproducible from the cached data using the provided scripts.

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